

MILANO MALPENSA, Italy

WMO: 160660

Lat: 45.631N

Lon: 8.728E

Elev: 767

StdP: 14.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	21.3	5.3	7.4	37.2	10.3	9.6	31.5	19.9	50.3	15.9	46.9	1.0	0	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	89.8	70.6	87.6	69.8	85.5	69.0	73.5	85.6	72.0	83.5	70.6	81.7	6.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	113.0	78.5	68.2	106.5	77.8	66.5	100.5	76.2	37.4	84.6	36.1	83.3	34.9	81.9	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	11.4	9.4	DB	12.4	93.2	4.1	2.4	9.4	94.9	7.0	96.4	4.6	97.7	1.6	99.5	(4)
			WB	11.3	75.6	4.0	2.5	8.5	77.4	6.2	78.9	3.9	80.3	1.0	82.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	36.1	38.7	46.6	53.6	61.8	69.6	73.5	72.7	64.6	55.6	45.3	37.3	(6)
(7)		DBStd	14.32	5.15	5.52	5.74	5.45	5.20	5.55	4.37	4.85	5.39	5.85	5.90	5.65	(7)
(8)		HDD50	1476	431	316	132	25	1	0	0	0	18	159	394	(8)	
(9)		HDD65	4507	896	735	570	342	126	19	2	4	71	292	591	859	(9)
(10)		CDD50	3199	0	0	28	134	367	587	730	704	438	193	18	0	(10)
(11)		CDD65	756	0	0	0	1	27	156	267	244	59	2	0	0	(11)
(12)		CDH74	7160	0	0	6	27	310	1527	2616	2202	452	20	0	0	(12)
(13)		CDH80	2271	0	0	0	4	48	462	916	762	77	2	0	0	(13)
(14)	Wind	WSAvg	3.3	2.6	3.0	3.8	4.0	4.0	3.9	3.8	3.7	3.2	2.7	2.8	2.4	(14)
(15)	Precipitation	PrecAvg	42.8	2.8	3.2	3.9	3.2	4.8	3.7	2.6	3.9	3.5	4.7	4.8	2.2	(15)
(16)		PrecMax	66.5	8.1	10.6	11.0	8.9	13.1	7.6	6.3	9.4	10.4	13.3	8.9	5.1	(16)
(17)		PrecMin	28.2	0.1	0.0	0.0	0.0	0.9	1.1	0.6	0.3	0.0	0.0	0.3	0.2	(17)
(18)		PrecStd	11.7	2.0	2.6	2.9	2.2	3.3	2.0	1.5	2.3	2.9	4.0	2.7	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	58.8	64.1	73.4	77.1	84.4	91.6	93.0	93.5	85.5	76.1	64.3	57.1	(19)	
(20)		MCWB	46.1	49.0	53.8	59.9	65.0	69.7	72.1	72.2	67.8	63.3	53.8	45.9	(20)	
(21)		DB	53.0	57.5	67.7	73.2	80.7	88.0	89.6	89.9	82.4	71.9	60.5	52.7	(21)	
(22)		MCWB	43.5	45.6	52.1	57.7	63.8	69.5	70.4	71.0	66.6	60.3	52.0	44.9	(22)	
(23)		DB	49.6	54.0	64.2	70.1	77.3	85.6	87.5	87.2	79.1	69.4	57.3	49.9	(23)	
(24)		MCWB	41.6	43.9	50.6	55.8	62.1	68.9	69.8	70.1	64.3	59.6	51.0	44.1	(24)	
(25)		DB	46.2	50.4	60.7	66.8	74.2	82.4	85.3	84.3	76.3	66.1	55.2	47.4	(25)	
(26)		MCWB	40.3	42.2	49.1	54.1	60.5	67.2	69.0	68.3	63.0	57.7	50.6	42.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	48.9	49.7	55.3	61.2	68.0	74.6	75.4	75.0	70.2	65.3	56.5	50.1	(27)	
(28)		MCDWB	55.7	60.3	67.7	74.0	79.0	87.2	86.7	87.0	80.3	70.2	60.5	51.2	(28)	
(29)		WB	45.1	47.4	53.4	58.6	65.1	72.1	73.5	73.3	68.4	63.1	54.8	47.3	(29)	
(30)		MCDWB	49.6	54.4	64.0	70.0	77.2	84.2	85.3	85.7	78.2	68.3	58.0	50.0	(30)	
(31)		WB	42.9	45.3	51.8	57.0	63.3	70.2	72.1	71.9	67.0	61.5	53.1	45.4	(31)	
(32)		MCDWB	46.8	51.0	61.0	67.4	74.6	82.0	83.4	83.3	76.2	66.4	56.1	48.2	(32)	
(33)		WB	40.9	43.5	50.1	55.3	61.8	68.4	70.7	70.3	65.3	59.8	51.4	43.5	(33)	
(34)		MCDWB	44.5	48.1	58.4	63.9	72.1	79.5	81.7	81.2	73.3	64.6	54.1	46.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	17.9	20.4	22.5	20.6	20.0	20.3	20.7	20.1	19.4	17.3	16.0	17.4	(35)	
(36)		5% DB	25.2	28.1	30.4	26.9	24.9	23.2	23.0	23.2	22.8	21.6	21.1	22.4	(36)	
(37)		MCWBR	16.8	18.2	17.0	13.5	11.3	9.8	9.0	8.5	9.6	11.7	13.8	15.4	(37)	
(38)		MCDBR	19.9	23.3	25.7	23.1	22.2	21.1	20.4	20.9	19.3	16.7	14.9	16.7	(38)	
(39)		5% WB	14.4	16.1	15.3	12.4	11.0	9.7	9.2	9.1	9.1	9.8	11.0	12.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.370	0.421	0.467	0.469	0.489	0.469	0.463	0.452	0.434	0.375	0.334	(40)	
(41)		taud	2.344	2.252	2.135	2.048	2.101	2.089	2.147	2.168	2.159	2.160	2.288	2.374	(41)	
(42)		Ebn at Noon	237	251	252	251	256	250	254	250	240	223	219	226	(42)	
(43)		Edh at Noon	25	33	42	50	49	50	47	44	41	36	26	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	465	724	1143	1427	1732	2002	2035	1726	1299	773	460	387	(44)	
(45)		RadStd	65	102	148	169	144	123	131	112	93	117	80	55	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.03	N/A	N/A	+1.67	N/A	N/A	N/A	-232	+265	+148	(46)
(47)	Regional (83 neighbors)	+0.58	N/A	N/A	+1.26	+0.76	+0.59	N/A	N/A	+104	+22	(47)

Nomenclature: See separate page